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- Internship : Internova AI Internship
- Week : Week 1 Submission
- Date : 09 August 2026

## Week 2 Assignment: Python for AI & Data Handling

### Student Performance Data Analysis

Objective: Learn how Python is used for data handling in Artificial Intelligence using NumPy, Pandas and Matplotlib.

#### Task 1: NumPy Fundamentals

NumPy is a Python library for numerical computing. It provides efficient multidimensional arrays and mathematical operations. This task demonstrates 1D and 2D arrays, indexing, slicing, shape, dimensions and basic statistical operations.

#### Task 2: Pandas & Dataset Handling

Pandas provides DataFrame objects for handling tabular data. The dataset was loaded from CSV, inspected using head(), tail(), shape, columns and dtypes, summarized with describe(), and queried using column selection and Boolean filtering.

#### Task 3: Data Cleaning

The raw dataset contained 4 missing cells and 1 duplicate record. Numerical missing values were filled with column medians, duplicate records were removed, and numeric/text columns were converted to suitable data types. After cleaning, missing cells were 0 and duplicate records were 0.

#### Task 4: Data Visualization

Four Matplotlib charts were created: a bar chart, line chart, histogram and scatter plot. These graphs make subject averages, student performance distribution, and relationships between study hours and marks easier to Understand.

### Task 5: Mini Project

The cleaned dataset contains 20 unique students. Average marks were 77.99, highest average was 97.67, lowest average was 48.33. Using 40 as the pass threshold, 20 students passed and 0 failed.

### Task 1 Output:

```
1D Array: [10 20 30 40 50]
2D Array:
[[10 20 30]
 [40 50 60]]

First element of 1D array: 10
2D array element [1,2]: 60
1D slice [1:4]: [20 30 40]
Shape of 2D array: (2, 3)
Dimensions of 2D array: 2

Mean: 30.0
Maximum: 50
Minimum: 10
Sum: 150

a1 + 5 = [15 25 35 45 55]
a1 * 2 = [ 20  40  60  80 100]
```

### Task 2 Output:

Csv:

H22

fx

Pass

|    | A          | B      | C         | D          | E     | F       | G       | H      | I | J | K |
|----|------------|--------|-----------|------------|-------|---------|---------|--------|---|---|---|
| 1  | Student_ID | Name   | Study_Hou | Attendance | Maths | Science | English | Result |   |   |   |
| 2  | 101        | Aarav  | 5         | 85         | 78    | 82      | 80      | Pass   |   |   |   |
| 3  | 102        | Diya   | 7         | 92         | 91    | 89      | 94      | Pass   |   |   |   |
| 4  | 103        | Rohan  | 3         | 68         | 55    | 61      | 58      | Pass   |   |   |   |
| 5  | 104        | Ananya | 8         | 95         | 88    | 92      | 90      | Pass   |   |   |   |
| 6  | 105        | Kabir  | 6         | 88         | 72    | 75      | 78      | Pass   |   |   |   |
| 7  | 106        | Isha   | 4         | 76         | 64    | 70      | 73      | Pass   |   |   |   |
| 8  | 107        | Vikram | 9         | 97         | 95    | 96      | 91      | Pass   |   |   |   |
| 9  | 108        | Meera  | 5         | 82         | 81    | 79      | 83      | Pass   |   |   |   |
| 10 | 109        | Arjun  | 2         | 61         | 48    | 52      | 55      | Fail   |   |   |   |
| 11 | 110        | Sneha  | 7         | 90         | 86    | 88      | 89      | Pass   |   |   |   |
| 12 | 111        | Rahul  |           | 74         | 67    |         | 69      | Pass   |   |   |   |
| 13 | 112        | Pooja  | 6         | 89         |       | 80      | 85      | Pass   |   |   |   |
| 14 | 113        | Kiran  | 4         | 79         | 73    | 77      | 75      | Pass   |   |   |   |
| 15 | 114        | Nisha  | 10        | 98         | 99    | 98      | 96      | Pass   |   |   |   |
| 16 | 115        | Aditya | 8         | 93         | 90    | 93      | 88      | Pass   |   |   |   |
| 17 | 116        | Tanya  | 3         |            | 59    | 62      | 64      | Pass   |   |   |   |
| 18 | 117        | Manav  | 5         | 84         | 76    | 74      | 79      | Pass   |   |   |   |
| 19 | 118        | Riya   | 7         | 91         | 84    | 86      | 90      | Pass   |   |   |   |
| 20 | 119        | Sahil  | 2         | 58         | 45    | 49      | 51      | Fail   |   |   |   |
| 21 | 120        | Neha   | 9         | 96         | 94    | 95      | 92      | Pass   |   |   |   |
| 22 | 103        | Rohan  | 3         | 68         | 55    | 61      | 58      | Pass   |   |   |   |
| 23 |            |        |           |            |       |         |         |        |   |   |   |
| 24 |            |        |           |            |       |         |         |        |   |   |   |
| 25 |            |        |           |            |       |         |         |        |   |   |   |
| 26 |            |        |           |            |       |         |         |        |   |   |   |
| 27 |            |        |           |            |       |         |         |        |   |   |   |
| 28 |            |        |           |            |       |         |         |        |   |   |   |
| 29 |            |        |           |            |       |         |         |        |   |   |   |
| 30 |            |        |           |            |       |         |         |        |   |   |   |

CSV import

First 5 rows:

| Student_ID | Name   | Study_Hours | Attendance | Maths | Science | English | Result |
|------------|--------|-------------|------------|-------|---------|---------|--------|
| 101        | Aarav  | 5.0         | 85.0       | 78.0  | 82.0    | 80      | Pass   |
| 102        | Diya   | 7.0         | 92.0       | 91.0  | 89.0    | 94      | Pass   |
| 103        | Rohan  | 3.0         | 68.0       | 55.0  | 61.0    | 58      | Pass   |
| 104        | Ananya | 8.0         | 95.0       | 88.0  | 92.0    | 90      | Pass   |
| 105        | Kabir  | 6.0         | 88.0       | 72.0  | 75.0    | 78      | Pass   |

Last 5 rows:

| Student_ID | Name  | Study_Hours | Attendance | Maths | Science | English | Result |
|------------|-------|-------------|------------|-------|---------|---------|--------|
| 117        | Manav | 5.0         | 84.0       | 76.0  | 74.0    | 79      | Pass   |
| 118        | Riya  | 7.0         | 91.0       | 84.0  | 86.0    | 90      | Pass   |
| 119        | Sahil | 2.0         | 58.0       | 45.0  | 49.0    | 51      | Fail   |
| 120        | Neha  | 9.0         | 96.0       | 94.0  | 95.0    | 92      | Pass   |
| 103        | Rohan | 3.0         | 68.0       | 55.0  | 61.0    | 58      | Pass   |

Task 3 Before/After Cleaning:

Cleand csv



```
BEFORE CLEANING
Missing values:
Student_ID    0
Name          0
Study_Hours   1
Attendance    1
Maths         1
Science       1
English       0
Result        0
```

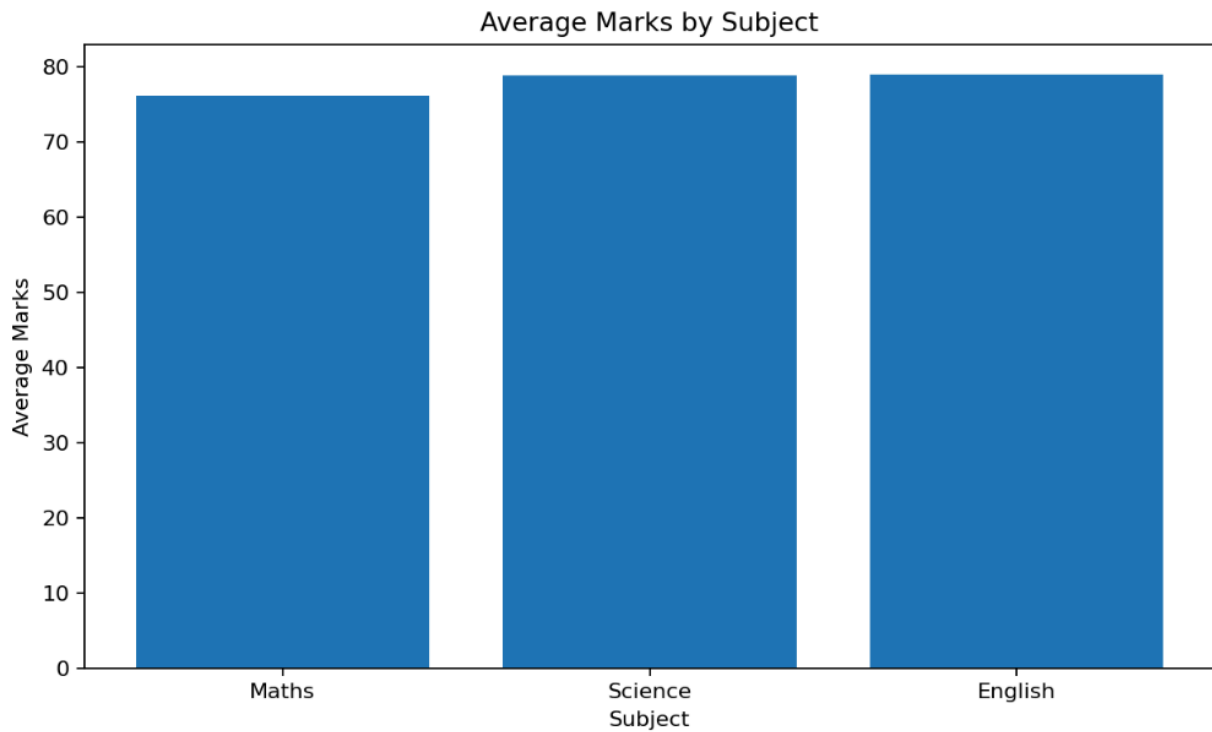
Duplicate records: 1

```
AFTER CLEANING
Missing values:
Student_ID    0
Name          0
Study_Hours   0
Attendance    0
Maths         0
Science       0
English       0
Result        0
```

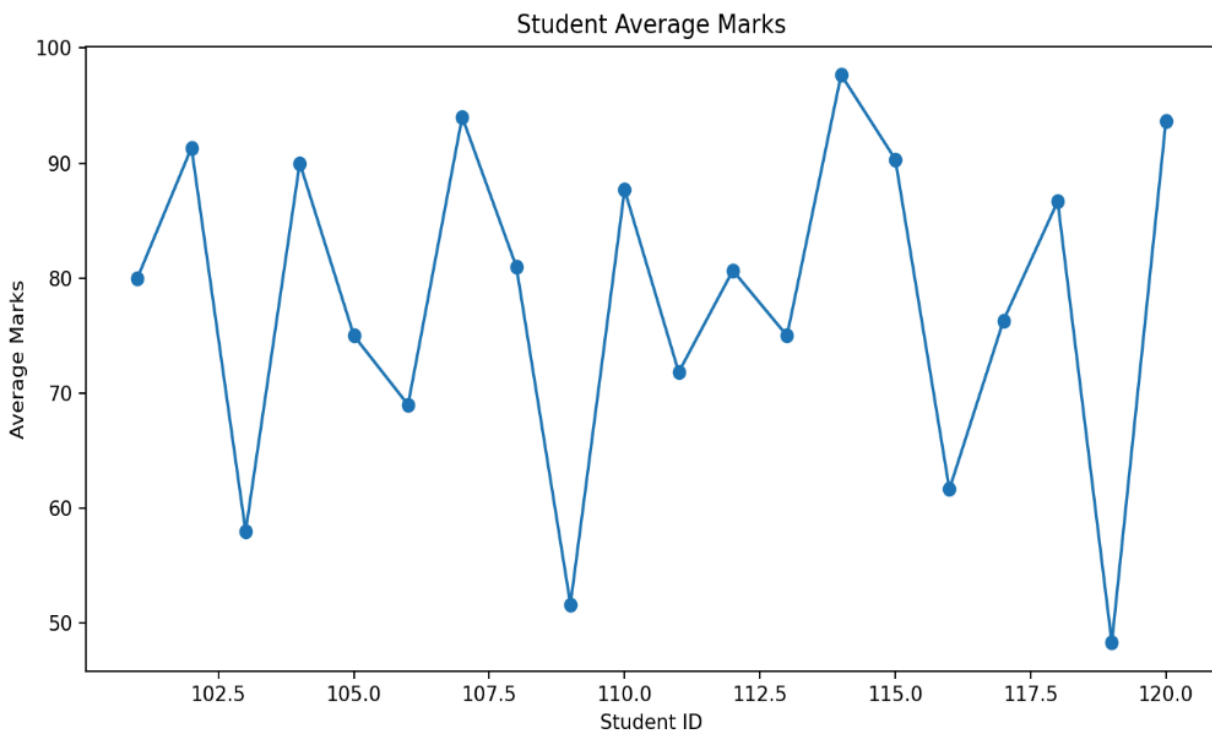
Duplicate records: 0

Cleaned rows: 20

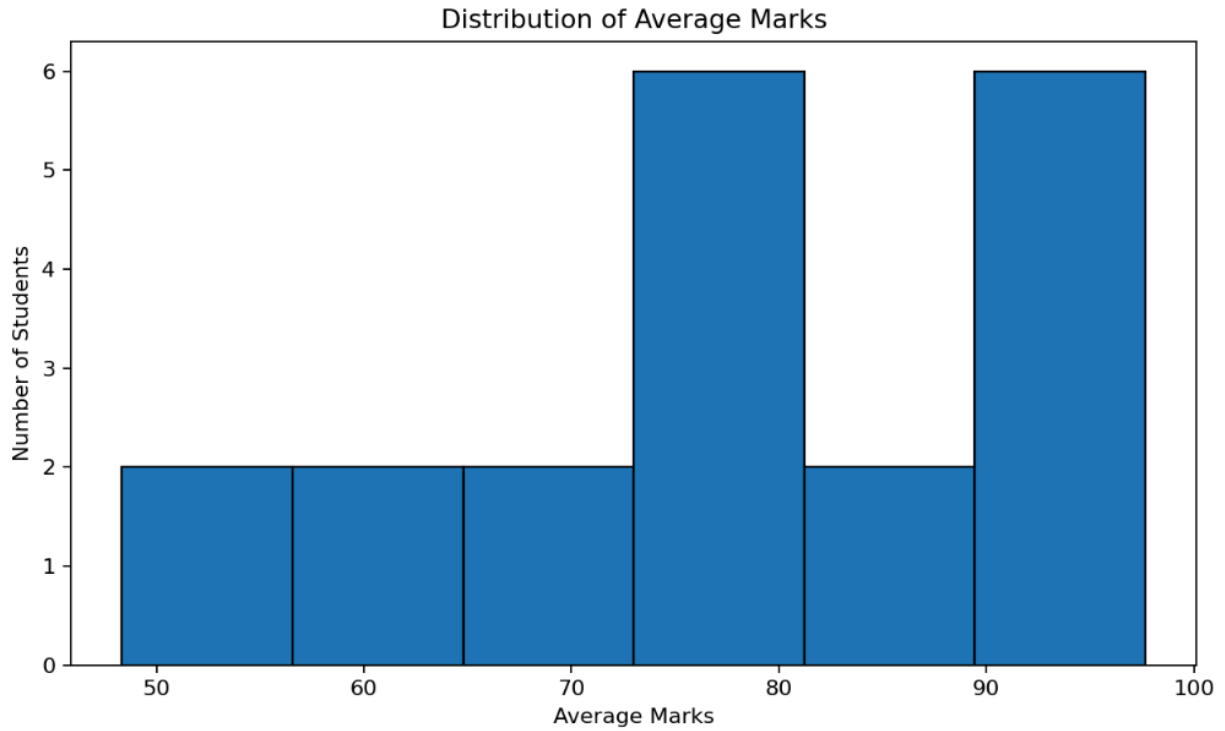
Bar Chart — Average marks by subject:



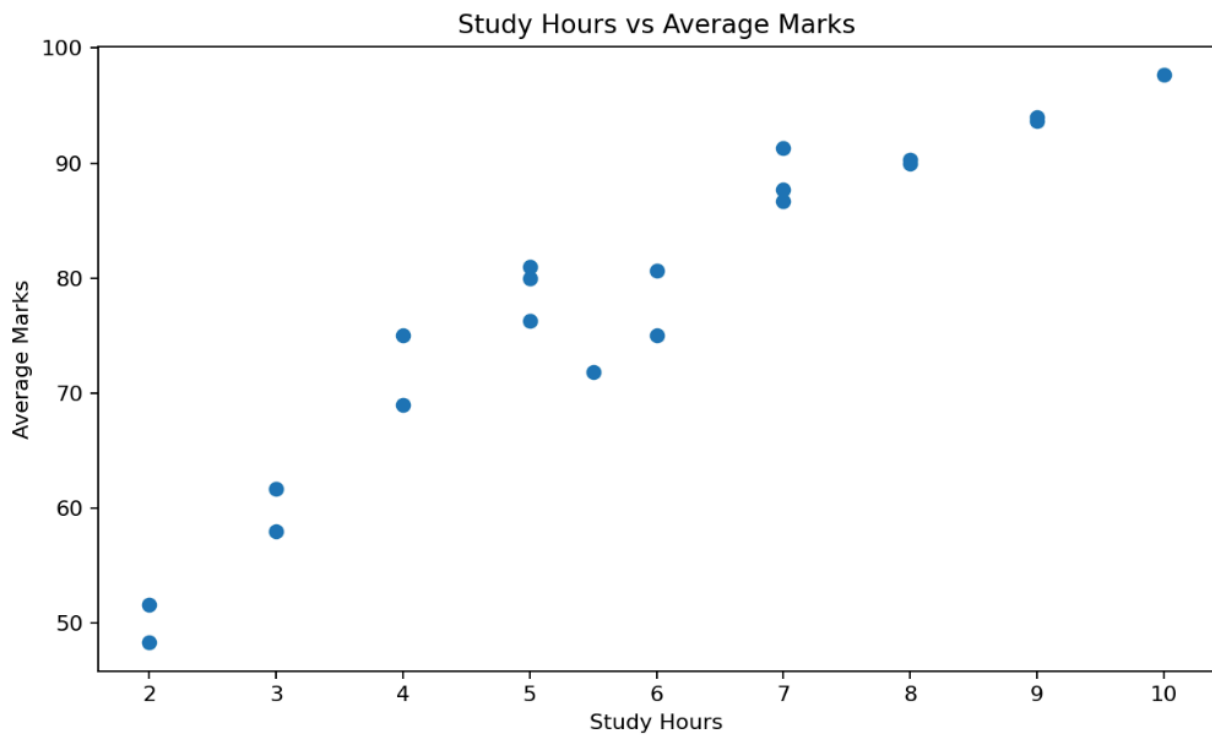
Line Chart — Student average marks:



Histogram — Distribution of average marks:



Scatter Plot — Study hours vs average marks:



Mini Project Conclusion

The student performance dataset provides a basic view of how study habits, attendance and subject marks relate to academic results. After cleaning the data, the dataset contained 20 unique student records. Missing numerical values were filled using the median of their respective columns, while the duplicate record was removed. The overall average mark across Maths, Science and English was 77.99. The highest student average was 97.67, while the lowest was 48.33. Using an average mark of 40 as the pass threshold, 20 students passed and 0 students failed. The visualizations show the distribution of student performance and allow comparisons between subjects. The scatter plot comparing study hours with average marks produced a correlation of 0.95, suggesting that study time has a relationship with performance in this sample, although correlation alone does not prove causation. Attendance can also be compared with marks to identify patterns. Overall, the analysis demonstrates how Pandas can clean and summarize data and how Matplotlib can turn numerical results into understandable visual information. These steps form a useful foundation for future machine-learning work.

#### Submission Checklist

- 3 Jupyter Notebook (.ipynb)
- 3 Python/analysis code included in notebook
- 3 Raw dataset (.csv)
- 3 Cleaned dataset (.csv)
- 3 Output screenshots
- 3 Four required visualizations
- 3 Mini project charts
- 3 PDF report